A computer-assisted proof of the final evolution of the Pythagorean three-body problem

Preprint (version 1.2, 2026-10-13; version 1: 2026-10-12). Internally reviewed; not yet externally peer reviewed and not formally (machine-)verified. Section 8 lists the remaining assumptions. Released on Zenodo (DOI in Section 9); not yet submitted to a journal. **Change in version 1.2:** the classical collision facts that version 1 cited for global existence through binary collisions are replaced by a self-contained proof for the planar case (Appendix A.2); see Section 8, item 2.

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The use of AI tools is described in the section "Use of AI tools". Every number is traceable to a file in the reproduction package (Section 9).

Abstract

In the Pythagorean (Burrau) three-body problem, point masses 3, 4 and 5 start at rest at the vertices of a 3-4-5 right triangle, each opposite the side of its own length. Numerical studies since Szebehely and Peters (1967) agree that the motion ends with the mass-3 body escaping and the masses 4 and 5 forming a bound binary. We describe a computer-assisted argument for this final evolution. A validated Taylor-series integrator with outward-rounded dyadic interval arithmetic (128-bit mantissas, order 20) and a Lohner–QR parallelepiped representation with a width-weighted pivoting rule encloses the exact solution on $t \in [0, 61.0055]$ in 31,811 accepted steps. From the stored enclosures at $t = 59.50, \ldots, 61.00$ we check a sufficient escape condition (★) — Yoshida's (1972) Theorem 1 with $d = m_1 m_2/K$, which is also a special case of the corrected form of his Corollary that Yoshida himself published (Erratum 1973; Criterion A′, 1974) — using two independently written exact/interval checkers. Both certify (★) on every stored box from $t = 59.73$ to $t = 61.00$ and on none before. Together with a hand proof of (★) and of the two potential-bound lemmas it needs, this yields that the mass-3 body escapes with $\rho(t) \geq \rho_0 + C(t - t_0)$ for some $C > 0$ and the 4–5 separation stays below 1200/769 for all later times, where binary collisions (if any) are continued by Levi-Civita regularization. We report failed attempts and lower precisions (112 and 96 bits do not reach $t = 61$) and state the limits of the verification: one-ulp rounding faults in the interval layer are excluded only by code reading, and no formal verification or external review has yet been done. Global existence through binary collisions is proved in an appendix from standard ODE theory (planar case). To our knowledge, no rigorous proof of this final evolution has been published; our literature search (Section 1.2) cannot exclude one.

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1. Introduction

1.1 The problem

Three bodies with masses $m_0 = 3$, $m_1 = 4$, $m_2 = 5$ (gravitational constant $G = 1$) start at rest at

$q_0 = (1, 3)$, $q_1 = (-2, -1)$, $q_2 = (1, -1)$,

so that the mutual distances are $|q_0q_1| = 5$, $|q_0q_2| = 4$, $|q_1q_2| = 3$. The centre of mass is at the origin, the total momentum and angular momentum are zero, and the energy is exactly

$$H = -(3\cdot 4/5 + 3\cdot 5/4 + 4\cdot 5/3) = -769/60.$$

The problem is usually attributed to Burrau (1913). Szebehely and Peters (1967) integrated it numerically with Levi-Civita regularization of close encounters and reported that the bodies 4 and 5 end as a binary while body 3 recedes indefinitely. Yoshida (1972) applied an analytic hyperbolic–elliptic escape criterion at $t = 59.72$ to numerical data that he attributes to Yuasa (1971, private communication; Yoshida 1972, p. 406 — no publication of these data was found in Crossref or a web search). Later studies are numerical: Aarseth et al. (1994) on the chaotic neighbourhood of the initial condition; Hirayama, Nishikawa and Shiraishi (2006) and Hirayama (2019) with high-order Taylor series methods, the latter in quadruple precision (title and bibliographic record checked; text not read); and the arbitrary-precision Bulirsch–Stoer code Brutus. Boekholt and Portegies Zwart (2015) needed a tolerance of 10^{-14} and an 88-bit word length for a solution converged to three decimal places and report dissolution at about $t = 60$. Portegies Zwart and Boekholt (2018) used a word length of 136 bits and a tolerance of at most 10^{-24} so that the time-reversed solution returns to the initial condition, and report a dissolution time $t \simeq 63.4$ defined as the time at which the linear behaviour of the phase-space separation overtakes its exponential behaviour (a fitted model); they note that other escape criteria give different times. This time is therefore not

comparable with the first time t ≈ 59.73 at which our sufficient condition (★) holds. These are converged numerical solutions, not error-bounded ones.

1.2 Prior rigorous work

Validated (interval) integration with Taylor methods and Lohner's QR wrapping-control is standard (Lohner 1988, 2001; Nedialkov, Jackson and Corliss 1999; Zgliczyński 2002; the CAPD library, Kapela et al. 2021), and computer-assisted proofs exist for other three-body questions — e.g. choreographies and the figure-eight orbit (Kapela and Zgliczyński 2003; Kapela and Simó 2007) and collision/near-collision orbits of the restricted problem (Capiński, Kepley and Mireles James 2023). None of the components of our integrator is new.

We searched ADS (abstracts and the citation list of Yoshida 1972), scite, the arXiv API, zbMATH Open (via its public API) and the general web (English, Japanese, Russian), with both supporting and opposing queries, for a rigorous proof of the Pythagorean final evolution. **We did not find one.** zbMATH Open returned, for this problem, only numerical studies (Aarseth et al. 1994; Janssens 2011; Rose and Dullin 2013; Hirayama 2019), and for interval methods in the three-body problem only works on other questions. This is a statement about our search, not a proof of absence: MathSciNet (subscription) was not searched, nor were thesis repositories, and validated-numerics groups were not contacted. **To our knowledge, no rigorous proof of the final evolution of the Pythagorean problem has been published.**

1.3 On Yoshida's Corollary

Yoshida's (1972) Corollary (p. 402) states an escape criterion without the hypothesis r(0) ≤ d, but its proof uses that hypothesis in its first step. **Yoshida himself corrected this, first in an Erratum (PASJ 25, p. 285, 1973)**, which replaces condition (26) of the Corollary by d ≥ r(0), and again in 1974 (PASJ 26, p. 369, Note, calling the 1972 conditions (25), (26) "incorrect" and replacing them by ρ(0) ≥ ad ≥ a r(0), "Criterion A′"). The same erratum also corrects inequality (40) on p. 405 (Theorem 2, which we do not use). We did not discover this gap; it was corrected by the author. Our project first noticed the gap independently (a state at t ≈ 17.118 on the Pythagorean trajectory satisfies the uncorrected hypotheses while violating the conclusion), and only afterwards found the author's corrections (the 1973 erratum was found last, while checking the references of this paper). Yoshida's own Pythagorean application (t = 59.72, r = 0.42 < d_1) satisfies the corrected conditions, so his 1972 numerical conclusion is not affected; it rests, however, on 1971 floating-point data without error bounds. We use only the corrected form (★) below, which requires r < d_1 explicitly.

1.4 Contribution and structure

We give (i) an interval enclosure of the exact solution on [0, 61.0055] (Sections 2, 4); (ii) a proof of the sufficient escape condition (★) with explicit lemmas (Section 3, Appendix A); (iii) an exact check of (★) on the enclosures by two independent programs (Section 4.2); (iv) a description of how the code was checked (Section 5) and of what remains unchecked (Section 8). All numbers, code hashes and run records are in a reproduction package (Section 9, `REPRODUCE.md`).

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2. Method

2.1 Interval arithmetic on dyadic numbers

All rigorous quantities are closed intervals [a, b] whose endpoints are dyadic rationals m·2^e with |m| < 2^p, where p is the working precision (128 bits in the main run). Each arithmetic operation is evaluated exactly in rational arithmetic (Python integers and `Fraction`) and the result is rounded **outward**: the lower endpoint down and the upper endpoint up to p bits. Square roots use integer square roots (`math.isqrt`) rounded outward. A single precision is fixed for the whole run; every state endpoint is checked to be dyadic with mantissa ≤ p bits at each step (otherwise the step stops with an error). This "single precision" rule was adopted after the interval layer's independent checker showed that mixed-precision inputs make `intersect` slightly widen its result (an issue of contract, not of containment; 0 containment violations).

2.2 Taylor coefficients

The equations of motion in Cartesian coordinates, $\ddot{q}a = \Sigma\{b\neq a\}\ m_b\ (q_b - q_a)/|q_b - q_a|^3$, are expanded in Taylor series by automatic differentiation; the factor |r|^(−3) = s^(−3/2) with s = |r|² uses the power recursion k s_0 $w_k = \Sigma\ ((\alpha+1)i - k)\ s_i\ w_{k-i}$, α = −3/2. The order is p = 20. The Taylor remainder is enclosed by evaluating the (p+1)-th coefficient over an a-priori enclosure of the trajectory.

2.3 One step: a-priori enclosure, Lohner–QR update

The state set is a parallelepiped X_k = { c + B r : r ∈ R }, with centre c (exact dyadic vector), frame B (exact dyadic 12×12 matrix) and interval vector R. One step of size h does:

1. **A-priori (Picard) enclosure.** Find a box P with $\hat{X}$ + [0, h]·F(P) ⊆ P, where $\hat{X}$ is the interval hull of the enclosure box of X_k and c. Successful evaluation of F(P) requires every pairwise squared distance to have a positive lower bound, so **P contains no collision configuration**; a Banach fixed-point argument then shows every solution starting in $\hat{X}$ stays in P on [0, h]. If the check fails, h is halved.
2. **Centre image.** T(c) + Z, with T(c) the order-p Taylor polynomial at c and Z = h^{p+1} [x_{[p+1]}](P).
3. **Jacobian enclosure.** [J] ⊇ Dφ_h(y) for all y ∈ $\hat{X}$, from the Taylor coefficients of the variational equation (forward-mode differentiation in 12 directions over $\hat{X}$) plus a remainder term using an a-priori enclosure W of the variational matrix. (The code review found that this remainder term is redundant for the correctness of the final set, because Z is already evaluated over P; it only makes the set slightly larger.)
4. **New centre** c′ = the p-bit floor of the midpoint of T(c) + Z.
5. **New frame.** B′ = Q from a column-pivoted Householder QR factorisation (double precision) of A = mid([J])·B; Q is converted exactly to a dyadic matrix. **Its inverse is computed exactly** in rational arithmetic (Gauss–Jordan on `Fractions`) and then enclosed outward. B′ need only be invertible for correctness; orthogonality only affects tightness.
6. **New label** R′ = [B′⁻¹]([J]B) R + [B′⁻¹](T(c) + Z − c′).

By the mean-value theorem on the convex set $\hat{X}$ and inclusion monotonicity, φ_h(X_k) ⊆ X_{k+1}. The step also returns, for any 0 < τ ≤ h, an enclosure of φ_τ(X_k) ("in-step box"). This is the classical Lohner scheme in the form of Nedialkov–Jackson–Corliss (1999) and the C¹-Lohner literature (e.g. Walawska and Wilczak 2016); we claim nothing new for it.

2.4 Width-weighted pivoting

With the textbook pivoting (largest remaining column norm of A) the run failed at the first close encounter (Section 6). We instead choose, at each QR stage, the remaining column maximising (remaining column 2-norm) × w(R_j), the column norm weighted by the width of the corresponding label component. Lohner recommends a *pivoted* QR factorisation for this purpose (as reported by Revol 2023, citing Lohner 2001). We have not verified whether Lohner's original pivot uses the same width weighting. Since correctness only needs B′ invertible (step 5), the pivot rule affects tightness only.

2.5 Step-size rule

Tolerance tol = 2^(−p+16)·max(1, ‖c‖∞). Trial step h = 2^⌊log₂ v⌋ with v = 0.9 (tol/m)^{1/20}, where m is the largest absolute endpoint of the order-20 coefficient at the centre; h ≤ 1/8. A step is rejected if the largest component width of Z exceeds tol, and retried with half the step, at most 20 times. The run stops if the enclosure width exceeds 1, if h < 2^{-60}, at t ≥ T_end, at the wall-clock limit, or on any exception or containment violation. The step rule affects efficiency only; correctness rests on Section 2.3.

2.6 Implementation

Pure Python 3 (integers, `fractions.Fraction`, `math.isqrt`), no external numerical library in the integrator. Files `dyad.py` (interval layer), `taylor_step.py` (Taylor coefficients, Picard enclosure), `lohner_step.py` (one Lohner step), `r3c_run.py` (driver), `r4_run.py` (thin wrapper: sets T_end = 61 and writes the enclosures at t = 59.50, …, 61.00 as exact fractions). A C++/GMP port of the Taylor core exists and was found bit-identical to the Python code on 23,452 test cases. **The results reported here come from the Python code**; the C++ core was used only for the full re-run R4R (Section 4.1), which reproduced them bit for bit.

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3. Escape criterion

3.1 Notation

Escaping body $m_0 = 3$; binary $m_1 = 4 \le m_2 = 5$; $M = m_0 + m_1 + m_2 = 12$. $\lambda = m_2/(m_1+m_2) = 5/9$, $\nu = m_1/(m_1+m_2) = 4/9$, $\mu = m_0(m_1+m_2)/M = 9/4$. $\bar{r} = q_2 - q_1$, $r = |\bar{r}|$. $G = (m_1q_1 + m_2q_2)/(m_1+m_2)$, $\bar{\rho} = q_0 - G$, $\rho = |\bar{\rho}|$, and $\dot{\boldsymbol{\rho}} = (\bar{\boldsymbol{\rho}} \cdot \dot{\bar{\boldsymbol{\rho}}})/\boldsymbol{\rho}$ (radial velocity, not speed). $K = -H = 769/60$, $d_1 = m_1m_2/K = 1200/769$, $\lambda d_1 = 2000/2307$. Define

$$W(\rho, d) = \lambda/(\rho + \nu d) + \nu/(\rho - \lambda d).$$

3.2 Statement

Theorem (★). Let $H < 0$ and $m_1 \le m_2$. If at some time t_0 the three strict inequalities

(C1) $r < d_1$, (C2) $\rho > \lambda\, d_1$, (C3) $\dot\rho > \sqrt{(2M \cdot W(\rho, d_1))}$

hold, then for all $t \geq t_0$ (with binary collisions of m_1, m_2, if any, continued by regularization): $r(t) < d_1$, $\dot\rho(t) > C := \sqrt{(\dot\rho_0^2 - 2M\, W(\rho_0, d_1))} > 0$, and $\rho(t) \geq \rho_0 + C(t - t_0)$. In particular body m_0 escapes and the binary stays bounded (hyperbolic–elliptic motion).

(★) is Yoshida's (1972) Theorem 1 (p. 400) with $d = d_1$, in which case its third condition coincides with (C3). It is also the case $d = d_1$ of the corrected Corollary (Erratum 1973: condition (26) replaced by $d \geq r(0)$) and of Yoshida's (1974) Criterion A′; A′ formally allows the boundary case $r(0) = d_1$, which cannot occur together with (C2), (C3) (Appendix A, Remark A.4.2). The 1974 proof of A′ was not read; we use only our own proof.

3.3 Lemmas

For $\lambda \geq \nu > 0$, $\lambda + \nu = 1$, $\rho > 0$, $r \geq 0$, $c = \cos\theta \in [-1, 1]$, let $r_1 = \sqrt{(\rho^2 + \nu^2 r^2 - 2\nu\rho r c)}$, $r_2 = \sqrt{(\rho^2 + \lambda^2 r^2 + 2\lambda\rho r c)}$, $P = \lambda/r_1 + \nu/r_2$, $F = \lambda(\rho - \nu r c)/r_1^3 + \nu(\rho + \lambda r c)/r_2^3 = -\partial P/\partial\rho$, $\hat F(\rho, d) = \lambda/(\rho + \nu d)^2 + \nu/(\rho - \lambda d)^2$.

Lemma A. If $d \geq 0$, $\rho > \lambda d$, $0 \leq r \leq d$, then $P(\rho, r, c) \leq W(\rho, d)$. **Lemma B.** Under the same hypotheses $F(\rho, r, c) \leq \hat F(\rho, d)$.

Both are equalities at $r = d$, $c = -1$. They correspond to Yoshida's (1972) Lemmas 1 and 2 ($n = 1$). Our proof (Appendix A) expands ρP and $\rho^2 F$ in Legendre polynomials, $\rho^2 F = \Sigma\,(n+1)\, s^n\, \beta_n\, P_n(c)$ with $s = r/\rho$ and $\beta_n = \lambda\nu^n + \nu(-\lambda)^n$, uses $(-1)^n\beta_n \geq 0$ (this is where $\lambda \geq \nu$ enters) and $|P_n| \leq 1$, and then monotonicity in s. The multipole expansion is a standard technique; we claim no novelty for the lemmas.

3.4 Proof of (★)

The full proof is given in Appendix A. Outline: a bootstrap argument on the closed set $\mathcal{G} = \{t : r \leq d_1$ and $\dot\rho \geq 0$ on $[t_0, t]\}$. Lemma B and integration against $\dot\rho \geq 0$ give $\tfrac{1}{2}\dot\rho^2 - MW(\rho, d_1) \geq \tfrac{1}{2}C^2$; Lemma A and the (vector) energy identity give $r \leq m_1 m_2/(K + \tfrac{1}{2}\mu C^2) < d_1$; continuity then contradicts $\sup \mathcal{G} < T^*$. On $\mathcal{G}$ the escaper stays at distance $\geq \delta = \rho_0 - \lambda d_1 > 0$ from both binary components. The statement (R) of Appendix A.2 then gives that the solution, continued through binary collisions by Levi-Civita regularization, exists for all time. (R) is **proved** in Appendix A.2 for the planar case from standard ODE theory: a separation bound keeps the regularized state in a compact set on finite time, and each unit of regularized time contributes a positive amount of physical time. (Version 1 cited classical facts here instead; see Section 8, item 2.)

3.5 Why the hypothesis $r < d_1$ matters (sanity checks)

Dropping any ingredient of (★) gives a criterion that is numerically falsified: with the factor 2 omitted, with λ and ν interchanged in W, or with (C1) removed, the internal review found states satisfying the modified condition after which body 3 returns (18, 19 and 1 numerically reliable examples, respectively). With (★) itself, 800 sampled states (400 random, 400 near the boundary) showed no violation of any conclusion up to $t = 40$. These are numerical tests, not proofs; they serve to detect sign or factor errors.

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4. Results

4.1 Enclosure on [0, 61] (run R4-A)

Quantity	Value
Precision, order, pivot	128 bits, $p = 20$, width-weighted
Final time reached	$t = 2047006249/33554432 \approx 61.005540$ (stop reason: target time reached)
Accepted steps / attempts / rejected	31,811 / 41,140 / 9,329
Step sizes	$2^{-25} \ldots 2^{-5}$
Largest component width of final enclosure	10^−9.545
Wall-clock time	37,015 s ≈ 10.3 h (one core of a Google Cloud n2-custom-12-12288 instance: 12 vCPU, 12 GiB memory, Debian; Python 3.13.5)
Containment cross-checks vs. a 60-digit reference trajectory, grid $t = k/100$, $k = 1 \ldots 6100$	path boxes P: 6,100 PASS; in-step boxes: 1,138 PASS, 4,955 not decidable (box narrower than 10^{-15}; reference inside); step-end boxes: 7 not decidable (reference inside). **Violations: 0**

The reference trajectory is a non-rigorous high-precision Taylor integration; the grid comparison is a consistency check that can expose bugs, not part of the proof. The closest encounter on the trajectory is between bodies 4 and 5 at $t \approx 15.830$ with separation $\approx 4.16 \times 10^{-4}$ (non-rigorous reference value).

The run stored, for each of the 151 grid times t_k = k/100, k = 5950, …, 6100, two enclosing boxes (302 boxes) with exact rational endpoints: a wide path box (component widths 1.1×10^{-2} to 6.3×10^{-2}) and a narrow in-step or step-end box (widths 4.8×10^{-13} to 3.3×10^{-9}).

Second run (R4-B, 144 bits). The same code at 144-bit precision also reached t = 61.000086 (52,388 accepted steps, 55,633 s, final largest width 10^−17.416, 0 violations). This is an independent *run*, not an independent *implementation*. Its stored boxes were **not** checked against (★). In its grid check, 800 in-step and step-end boxes at $t \le 8.13$, all narrower than 10^−28.9, did not contain the reference point; these are below the 10^{-15} decision threshold and counted as undecidable, not as violations. We attribute this to the limited accuracy of the reference trajectory itself (its nominal tolerance is 10^{-35} and errors grow through close encounters); **this explanation has not been checked**.

Bit-for-bit reproduction of the whole computation (R4R). The 128-bit computation on [0, 61] was re-run from the start with the Taylor-coefficient core replaced by the C++/GMP implementation of Section 2.6 (checked bit-identical to the Python core in 23,452 cases). All 31,811 steps agree in t, step size, number of rejected attempts and enclosure width (0 mismatches); the stored-box file and the grid-check file have the same SHA-256 as in the original run (`f65101c1…`, `fe53ded8…`); stop reason, final time and grid counts are identical. Wall-clock time was 7,974 s (2.2 h) instead of 37,015 s. **Limitation:** this is a second implementation of the same algorithm — the step rule, the QR frame update, the exact inverse and the containment checks are the same Python code — not an independent algorithm.

4.2 Escape check (Y1)

Two programs written for this purpose, sharing no code with each other or with the integrator, evaluated (C1)–(C3) on all 302 boxes: `y1_exact` (exact rationals; square roots bracketed by integer square roots to 2^{-200}) and `y1_mpiv` (mpmath interval arithmetic, 256 bits). A box passes only if each inequality holds at **every** point of the box (outer bounds). Exact constants were used (K = 769/60, d_1 = 1200/769).

Result	Value
Boxes passing in both checkers	256 = all boxes at all 128 grid times t = 59.73, …, 61.00
Boxes passing at $t \le 59.72$	0 of 46
Disagreements between checkers	0 of 302
Guaranteed minimum margins ($t \ge 59.73$)	C3: 0.0235 (wide box), 0.0374 (narrow box); C1: 0.46; C2: 0.85
Narrow box, t = 59.73: certified C3 margin	[0.03744808107650, 0.03744808108289] (reference value 0.03744808107969 inside)
Self-tests	253 checks, 0 failures; 8 deliberately wrong checkers (λ↔ν, wrong d_1, wrong M, (C1) dropped, …) all detected

The non-rigorous reference trajectory first satisfies (★) at $t \approx 59.7223$; the certified boxes switch from "fail" (narrow-box C3 margin −0.012143 at t = 59.72) to "pass" (+0.037448 at t = 59.73), consistent with this. As a further consistency check, the exactly known conserved quantities (momentum 0, angular momentum 0, energy −769/60) lie inside the corresponding interval evaluations of all 302 boxes.

4.3 Conclusion

Combining 4.1, 4.2 and (★): at t_0 = 59.73 the exact state satisfies (C1)–(C3), hence for all $t \ge 59.73$ the separation of bodies 4 and 5 stays below 1200/769 ≈ 1.5605 and body 3's distance from the binary's centre of mass grows at least linearly. On [0, 61.0055] the enclosures also show that no collision occurs. This conclusion is **conditional on the items in Section 8**.

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5. Verification of the implementation

Implementation, checking and review were done in separate AI-tool sessions with restricted file access, coordinated by the author (see "Use of AI tools" below). The "independence" in this section therefore means separately written code under a sealed written contract, run and re-run separately — not separate human teams.

Check	What was done	Outcome	Source
Interval layer (R0)	Independent checker: 200,000 random cases at precisions 24/53/128/300, boundary cases, sqrt; checked against exact rational values	Containment violations 0 (S1 477,198 checks, S2 434,956, S3 17,000); checker caught 25/25 planted faults	`rigorous/R0_FINAL_RUN_KO.md`
Taylor step (R1)	Independent checker vs. 130-digit references: oscillator, circular Kepler, Pythagorean single steps, path boxes	1600/1600/9/3200 pass, 0 violations; 19/19 planted faults caught	`rigorous/R1_FINAL_RUN_KO.md`
Lohner step, checker v1	Independent checker on both pivot rules	0 violations, but 10 cases (S2: 8, X4: 2) were refused because the *checker* supplied non-dyadic start states	`GATE_Y1_PLAN_KO.md` §6; `r3c-c`

Check	What was done	Outcome	Source
			`heck-v2-handover` note
Lohner step, checker v2	Fixed input generation (dyadic grid)	Width-pivot rule (used for R4): **PASS, 0 violations, 0 refused** (S1 160, S2 2,356, C 4,098, X4 732 cases). Colnorm rule: 0 violations, unresolved (52 cases refused/not run because boxes grew to contain a collision configuration)	`rigorous/R3C_CHECK_V2_FINAL_KO.md`
Code review	Line-by-line review of all five files against a stated contract, plus attack tests and mutation tests against independent mpmath references: 806 one-step cases (oscillator 480, Kepler 216, three-body 60, checker-gap class 50), 47 unit tests, 4 chains (incl. the Pythagorean problem with the R4 settings, t = 0 → 2.0005, 1,309 steps through the first close encounter, 219,912 set-membership checks)	**0 violations**; no path found by which an output box could miss the true solution; 3 issues unrelated to rigour (one robustness, two documentation). Exact inverse step confirmed. Of 12 mutants, 10 were detected; not detected: moving the exact inverse 1 ulp inward, and an exact-midpoint centre (which affects only tightness)	`integrator_review/INTEGRATOR_REVIEW_KO.md` §0, §3, §4-1, §4-2, §6
Determinism	Server run vs. earlier local diagnostic run	First 4,494 steps identical in time, step size, rejections and exact width	`GATE_R4_PLAN_KO.md` §2
Full re-run (R4R)	t = 0 → 61 re-run with the C++ Taylor core	31,811/31,811 steps identical; box and grid files SHA-256 identical; 2.2 h vs 10.3 h	`R4R_FINAL_JUDGMENT_KO.md`
C++ port (FC1)	Bit-identity check of a GMP port of the Taylor core	23,452 cases identical; used only for the R4R re-run, not for the proof record	`FC1_FINAL_JUDGMENT_KO.md`
(★) proof	Independent line-by-line review; new analytic proof of Lemmas A, B; sympy (33/33 identities, 4/4 negative controls); adversarial search of the lemmas (40,000 points, 50 digits); dynamic falsification (800 states)	Logic holds; three presentation fixes required (vector form of the energy identity, citation of (R), explicit definition of the bootstrap set)	`proof_review/PROOF_REVIEW_KO.md`
Global existence (R), v1.2	sympy check of the 10 identities used in Appendix A.2 (conservation laws of the regularized system, recovery of Newton's equations and of (E)); 4 deliberately wrong variants; non-rigorous comparison of the regularized and the Cartesian integration of the Pythagorean problem to t = 30	10/10 identities hold; 4/4 wrong variants rejected; positions agree to 7.5×10^{-8} through a 4–5 approach of 0.0022	`proof_review/s3/`
Literature	Direct reading of Yoshida 1972 (pp. 391–407), 1973 Erratum (p. 285) and 1974 (pp. 367–369) scans; zbMATH Open API search	Corollary gap confirmed and found corrected by the author (1973, 1974)	`LITERATURE_RECHECK_KO.md`, `LITERATURE_RECHECK_ZBMATH_KO.md`, this paper

6. What did not work

We report these because they bound the method.

Attempt	Outcome	Source
Axis-aligned boxes (R2), 128 and 256 bits	Enclosure width exceeded 1 at t = 6.968 and 7.055	`rigorous/R2_RESULT_KO.md`; r2-run note
Lohner–QR with textbook column-norm pivoting (R3c), 128 bits	Width exceeded 1 at t = 1.8767, just before the first close encounter (t ≈ 1.879, separation ≈ 0.0097). The pivot order changed at every step during the encounter. Changing only the pivot rule to the width-weighted one passed the same encounter at width ≈ 10^{-29}	`rigorous/R3C_RESULT_KO.md` §9-1, §9-3
Width pivot, 112 bits (R4-C)	Width exceeded 1 at t ≈ 54.498 (15,764 steps, 21,613 s); did not reach the escape window	`r3c_final_112w.json` in `r4_out.tgz`
Width pivot, 96 bits (R4-D)	Stopped at t ≈ 27.789: the a-priori box could not exclude a collision configuration (5,102 steps, 9,110 s)	`r3c_final_96w.json` in `r4_out.tgz`

None of these runs produced a containment violation in its grid checks; they failed by losing tightness, not by producing wrong boxes. So 128 bits is, for this step rule and order, close to the minimum that reaches t = 61 (two data points below; not a proof of minimality). The width growth over [0, 61] is roughly exponential and dominated by close encounters; we have not quantified the contribution of each encounter.

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7. Discussion

- **Cost.** About 10 h of single-core pure-Python time for the main run, under one second for the escape check. The method is entirely standard; the problem-specific choices are the width-weighted pivot, the precision, and stopping at t = 61 where the (★) margin is ≈ 0.93 (non-rigorous) instead of the first time it holds.
- **Why no regularization.** The closest approach ($\approx 4 \times 10^{-4}$) was passed in Cartesian coordinates; the a-priori enclosure certifies the absence of collisions on [0, 61.0055]. Regularization enters only through the theorem (★) for t > 61.
- **Comparison with Yoshida (1972).** His application used data with ρ, r differing by 1–6 % from our reference at t = 59.72; our certified escape time 59.73 is consistent with his 59.72.

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8. Limitations (what this paper does not establish)

1. **One-ulp rounding faults.** The independent checkers detect containment errors larger than their resolution; a fault that rounds 1 ulp inward (2^{-128} relative) is excluded only by code reading. Python integer, `Fraction`, `>>`, `math.isqrt` and `Fraction(float)` semantics are taken from the language specification.
2. **Global existence (R) — proved in version 1.2, not reviewed.** Version 1 cited the classical collision facts (Painlevé, Sundman, Levi-Civita, Kustaanheimo–Stiefel, Siegel–Moser) for the statement that binary collisions of bodies 4 and 5 can be continued and do not accumulate in finite time, and a verbatim statement for L = 0 under the separation bound (SEP) was not found. Version 1.2 replaces this by a self-contained proof (Appendix A.2) that uses only the existence, uniqueness and extension theorems for ODEs. It covers the planar problem, which is the case of the application; the spatial case is not treated. This proof is internal and has not been externally reviewed.
3. **No formal verification.** Neither the integrator nor the proof of (★) has been machine-checked.
4. **No external review.** All reviews so far were internal to the project.
5. **Literature.** Absence of a prior rigorous proof is a result of a bounded search (Section 1.2); **MathSciNet was not searched** (subscription), and zbMATH Open only through its public API. Lemma proofs in Yoshida (1972) were not audited step by step; the 1974 paper was read only on pp. 367–369. Hirayama (2019) and Lohner's thesis (1988) were identified from bibliographic records; their texts were not read.
6. **Reference trajectory.** The 144-bit grid anomaly in Section 4.1 is unexplained beyond a plausible inference.
7. **One integrator algorithm.** The 128- and 144-bit runs share all code. The R4R re-run reproduces the 128-bit computation bit for bit with a second implementation of the Taylor core, but the rest of the step (step rule, QR frame, exact inverse, containment checks) is the same code. Independence lies in the checkers and the review, not in a second integrator algorithm.

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9. Reproducibility

All code, plans (sealed by SHA-256 before each run), run records and the server archive are listed with hashes in `paper/REPRODUCE.md`. The main run needs about 10 h on one core; the escape check takes seconds. The package is released on Zenodo: https://doi.org/10.5281/zenodo.23317644 (version 1); all versions are listed under the concept DOI https://doi.org/10.5281/zenodo.23317643. Version 1.2 adds `proof_review/s3/` (the proof note for Appendix A.2 and its checks).

The released files are redacted copies. The originals are kept unchanged, and all seal hashes refer to them. In the copies, only the operating-system user name, the cloud instance name and project ID, and any e-mail address are replaced, by exact string replacement, with fixed placeholders (e.g. `/home/USER`). Numbers, fractions and computed values are never touched. A public table `REDACTION_MAP.json` lists, for every file, the SHA-256 of the original and of the copy and the number of replacements per rule. The script `redact_verify.py` (standard library only) checks that applying the inverse replacement to each copy reproduces the original byte for byte. It also checks that no placeholder occurred in the originals, which makes the inverse unambiguous. The output of this check is released with the package. Outside readers can verify the copy hashes against the table; the byte-level comparison requires the originals and is performed by the author.

Use of AI tools

- The design decisions and all final judgments (what to compute, acceptance criteria, which results to accept, what to claim) were made by the author.
- Code writing, the independent checkers, the code and proof reviews, the literature searches and the writing of the documents (including this paper) were carried out by an AI tool (Anthropic Claude), in multiple sessions kept separate from one another.
- Verification structure: each computational step followed a written plan sealed by SHA-256 before it was run; implementation and checking were assigned to separate sessions (a checker session did not read the code it checked, and vice versa); and a separate

coordinating session, directed by the author, independently re-ran key steps (e.g. the checker v2 final run, the FC1 checks and the Y1 escape check on freshly uploaded files).
- No AI tool is listed as an author. Responsibility for the content rests with the author.

Acknowledgements

Computations were run on Google Cloud (free trial credits).

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Appendix A. The escape criterion (★): statement and self-contained proof

Draft for a paper appendix, prepared 2026-10-12. It has not yet been reviewed externally or verified formally (e.g. in a proof assistant).

A.0 Scope and attribution

The criterion proved here is the case $d = m_1 m_2 / K$ of **Theorem 1 of Yoshida (1972)** [Y72, p. 400]. It is also the case $d = m_1 m_2 / K$ of **criterion A′ of Yoshida (1974)** [Y74, p. 369]; see Remark A.4.2.

The corollary to Theorem 1 in [Y72, p. 402] does not assume $r(0) \le d$. **Yoshida himself corrected this, first in an erratum [Y73] and then in [Y74, p. 369, Note].** The erratum [Y73] replaces condition (26) of the corollary by $d \ge r(0)$; it also corrects inequality (40) on p. 405 of [Y72]. The later Note [Y74, p. 369] calls conditions (25)–(26) of [Y72] "incorrect" and replaces them by $\rho(0) \ge ad \ge a\, r(0)$. The statement below assumes $r(t_0) < d_1$ explicitly, so that issue does not affect it.

Lemmas A and B below are the case $n = 1$ of Lemma 1 of [Y72, p. 393] and Lemma 2 of [Y72, p. 396]. Yoshida proves them by induction on n and by analysing the θ-derivative, respectively. We give different, self-contained proofs based on the Legendre (multipole) expansion. This is a standard technique, and we do not claim originality for it.

Everything below is proved in full for the planar problem, using standard ODE theory and elementary analysis. Global existence through binary collisions is proved in A.2 (new in version 1.2; version 1 cited classical collision facts there). The spatial case of A.2 is not treated; A.1, A.3 and A.4 hold in $\mathbb{R}^2$ and $\mathbb{R}^3$.

A.1 Setting and notation

Three point masses $m_0, m_1, m_2 > 0$ move in $\mathbb{R}^2$ or $\mathbb{R}^3$ under Newtonian gravity with $G = 1$. We work in the barycentric frame, where the total momentum is zero. The body m_0 is the candidate escaper. The other two are labelled so that

$$m_1 \le m_2.$$

Define

$$M = m_0 + m_1 + m_2, \qquad \lambda = \frac{m_2}{m_1 + m_2}, \qquad \nu = \frac{m_1}{m_1 + m_2}, \qquad m = \frac{m_1 m_2}{m_1 + m_2}, \qquad \mu = \frac{m_0(m_1 + m_2)}{M}.$$

Hence $\lambda \ge \nu > 0$, $\lambda + \nu = 1$, $\mu M \lambda = m_0 m_2$ and $\mu M \nu = m_0 m_1$.

Jacobi vectors.

- $\mathbf{r} = \mathbf{x}_2 - \mathbf{x}_1$.
- $\boldsymbol{\rho} = \mathbf{x}_0 - \mathbf{g}$, where $\mathbf{g} = (m_1\mathbf{x}_1 + m_2\mathbf{x}_2)/(m_1 + m_2)$.
- Then $\mathbf{x}_2 - \mathbf{g} = \nu\mathbf{r}$ and $\mathbf{x}_1 - \mathbf{g} = -\lambda\mathbf{r}$. Thus the lighter body m_1 is the one farther from $\mathbf{g}$.
- Write

$$r = |\mathbf{r}|, \quad \rho = |\boldsymbol{\rho}|, \quad r_1 = |\mathbf{x}_0 - \mathbf{x}_2| = |\boldsymbol{\rho} - \nu\mathbf{r}|, \quad r_2 = |\mathbf{x}_0 - \mathbf{x}_1| = |\boldsymbol{\rho} + \lambda\mathbf{r}|.$$

- For $r > 0$ let $c = \cos\theta = \boldsymbol{\rho}\cdot\mathbf{r}/(\rho r)$. Then

$$r_1^2 = \rho^2 + \nu^2 r^2 - 2\nu\rho r c, \qquad r_2^2 = \rho^2 + \lambda^2 r^2 + 2\lambda\rho r c.$$

Radial velocity. $\dot\rho := \dfrac{d}{dt}|\boldsymbol{\rho}| = \dfrac{\boldsymbol{\rho}\cdot\dot{\boldsymbol{\rho}}}{\rho}$. This is the radial component, *not* the speed $|\dot{\boldsymbol{\rho}}| \ge |\dot\rho|$.

Energy. In the barycentric frame, and with **vector** velocities,

$$H = \tfrac{1}{2}m|\dot{\mathbf{r}}|^2 + \tfrac{1}{2}\mu|\dot{\boldsymbol{\rho}}|^2 - \frac{m_1 m_2}{r} - \mu M\Big(\frac{\lambda}{r_1} + \frac{\nu}{r_2}\Big). \tag{E}$$

We assume $H < 0$ and put $K = -H > 0$ and $d_1 = m_1 m_2 / K$.

Equation of motion of $\boldsymbol{\rho}$.

$$\ddot{\boldsymbol{\rho}} = -M\Big(\frac{\lambda}{r_1^3} + \frac{\nu}{r_2^3}\Big)\boldsymbol{\rho} + \lambda\nu M\Big(\frac{1}{r_1^3} - \frac{1}{r_2^3}\Big)\mathbf{r} =: \Phi(\boldsymbol{\rho}, \mathbf{r}). \tag{EM}$$

Radial acceleration. Where $\rho > 0$, Lagrange's identity gives

$$\ddot\rho = \frac{\boldsymbol{\rho}\cdot\ddot{\boldsymbol{\rho}}}{\rho} + \frac{|\boldsymbol{\rho}|^2|\dot{\boldsymbol{\rho}}|^2 - (\boldsymbol{\rho}\cdot\dot{\boldsymbol{\rho}})^2}{\rho^3} \;\ge\; \frac{\boldsymbol{\rho}\cdot\ddot{\boldsymbol{\rho}}}{\rho}. \tag{RA}$$

The dropped term equals $|\boldsymbol{\rho}\times\dot{\boldsymbol{\rho}}|^2/\rho^3 \ge 0$. Nothing is assumed about the angular momentum; in particular $L = 0$ is allowed.

Auxiliary functions. For $\rho > 0$, $r \ge 0$ and $c \in [-1, 1]$ with $r_1, r_2 > 0$, set

$$P(\rho, r, c) = \frac{\lambda}{r_1} + \frac{\nu}{r_2}, \qquad F(\rho, r, c) = \frac{\lambda(\rho - \nu r c)}{r_1^3} + \frac{\nu(\rho + \lambda r c)}{r_2^3} = -\partial_\rho P.$$

- The potential term in (E) is $\mu M P$.
- By (EM), $\boldsymbol{\rho}\cdot\ddot{\boldsymbol{\rho}}/\rho = -M\,F(\rho, r, c)$.
- For $r = 0$ these reduce to $P = 1/\rho$ and $F = 1/\rho^2$ whatever the value of c.

For $d \ge 0$ and $\rho > \lambda d$, set

$$W(\rho, d) = \frac{\lambda}{\rho + \nu d} + \frac{\nu}{\rho - \lambda d}, \qquad \hat F(\rho, d) = \frac{\lambda}{(\rho + \nu d)^2} + \frac{\nu}{(\rho - \lambda d)^2} = -\partial_\rho W(\rho, d).$$

Then $W(\rho, d) = P(\rho, d, -1)$ and $\hat F(\rho, d) = F(\rho, d, -1)$. Geometrically, this is the collinear configuration in which the lighter component m_1, at distance λd from $\mathbf{g}$, lies between $\mathbf{g}$ and m_0.

A.2 Continuation through binary collisions and global existence (planar case)

Revised in version 1.2. Version 1 cited classical facts here (Painlevé, Sundman, Levi-Civita, Kustaanheimo–Stiefel, Siegel–Moser). This section proves what the proof of (★) needs. It uses only the existence, uniqueness and extension theorems for ODEs with a locally Lipschitz right-hand side [Ha, Ch. II]. The motion is planar. Plane vectors are identified with complex numbers, and $\langle a,b\rangle = \operatorname{Re}(\bar a b)$ is the dot product. The spatial case would need the Kustaanheimo–Stiefel analogue [KS1965]. It is not treated, and the application in A.5 does not need it.

The regularized system. Let $k = m_1 + m_2$. The relative motion satisfies $\ddot{\mathbf{r}} = -k\mathbf{r}/r^3 + \mathbf{f}$, where

$$\mathbf{f}(\boldsymbol{\rho},\mathbf{r}) = m_0\Big[\frac{\boldsymbol{\rho} - \nu\mathbf{r}}{r_1^3} - \frac{\boldsymbol{\rho} + \lambda\mathbf{r}}{r_2^3}\Big].$$

Following Levi-Civita [LC1920], use the state $Y = (t, u, w, h, \boldsymbol{\rho}, \boldsymbol{\sigma}) \in \mathbb{R}\times\mathbb{C}^2\times\mathbb{R}\times\mathbb{C}^2$ and write $z = u^2$ in place of $\mathbf{r}$. Consider, on the open set

$$D = \{Y :\ |\boldsymbol{\rho} - \nu z| > 0,\ |\boldsymbol{\rho} + \lambda z| > 0\},$$

the system (with $' = d/ds$)

$$\begin{aligned} t' &= |u|^2, & u' &= w, & w' &= \tfrac{h}{2}u + \tfrac{1}{2}|u|^2\,\bar u\,\mathbf{f}(\boldsymbol{\rho},z),\\ h' &= \langle 2uw, \mathbf{f}(\boldsymbol{\rho},z)\rangle, & \boldsymbol{\rho}' &= |u|^2\boldsymbol{\sigma}, & \boldsymbol{\sigma}' &= |u|^2\,\Phi(\boldsymbol{\rho},z). \end{aligned} \tag{LC}$$

Its right-hand side is real-analytic on D; the only denominators are r_1^3 and r_2^3. A binary collision $u = 0$ is an ordinary point of (LC).

Lemma A.2.1 (identities). Along any solution of (LC) in D:

1. $I := 2|w|^2 - h|u|^2$ is constant.
2. $\mathcal{H} := mh + \tfrac{1}{2}\mu|\boldsymbol{\sigma}|^2 - \mu M P(\boldsymbol{\rho},z)$ is constant.

Suppose in addition that $I = k$ and $u \neq 0$ on an s-interval. Let dots denote $d/dt = |u|^{-2}\,d/ds$. Then on that interval:

3. $\dot z = 2w/\bar u$ and $\ddot z = -kz/|z|^3 + \mathbf{f}$;
4. $h = \tfrac{1}{2}|\dot z|^2 - k/|z|$;
5. $\dot{\boldsymbol{\rho}} = \boldsymbol{\sigma}$ and $\ddot{\boldsymbol{\rho}} = \Phi$;
6. $\mathcal{H}$ is the energy (E).

The Jacobi forms of Newton's equations used here ($\ddot{\mathbf{r}} = -k\mathbf{r}/r^3 + \mathbf{f}$ and (EM)) follow from the Cartesian equations.

Proof. All items are polynomial or rational identities.

Item 3 by hand:

- $\dot z = 2uw/|u|^2 = 2w/\bar u$.
- $\frac{d}{ds}\dot z = 2w'/\bar u - 2|w|^2/\bar u^2$.
- Using $2|w|^2 = k + h|u|^2$,

$$\ddot z = |u|^{-2}\big(hu/\bar u + |u|^2\mathbf{f} - (k + h|u|^2)/\bar u^2\big) = \mathbf{f} - k/(|u|^2\bar u^2) = \mathbf{f} - kz/|z|^3.$$

Item 1:

- $\frac{d}{ds}2|w|^2 = 2h\langle w,u\rangle + 2|u|^2\langle w, \bar u\mathbf{f}\rangle$.
- $\frac{d}{ds}(h|u|^2) = |u|^2\langle 2uw, \mathbf{f}\rangle + 2h\langle u, w\rangle$.
- These are equal, since $\langle w, \bar u\mathbf{f}\rangle = \langle uw, \mathbf{f}\rangle$.

All six items, and the Jacobi forms of Newton's equations, were also checked by computer algebra (A.6). □

Definition (continued solution). Let the classical solution be given near t_0, with $r(t_0) > 0$. Choose u_0 with $u_0^2 = \mathbf{r}(t_0)$. Put $w_0 = \bar u_0\dot{\mathbf{r}}(t_0)/2$ and $h_0 = \tfrac{1}{2}|\dot{\mathbf{r}}(t_0)|^2 - k/r(t_0)$; then $I = k$.

The *continued solution* is the maximal solution $Y(s)$, $s \in [0, S^*)$, of (LC) in D with $Y(0) = (t_0, u_0, w_0, h_0, \boldsymbol{\rho}(t_0), \dot{\boldsymbol{\rho}}(t_0))$. It is read in physical time through $t(s)$. Let

$$T^* := \lim_{s \to S^*} t(s) \in (t_0, \infty].$$

Then $[t_0, T^*)$ is its maximal interval of existence.

The choice $-u_0$ gives $(t, -u, -w, h, \boldsymbol{\rho}, \boldsymbol{\sigma})$, which is the same motion. Physically, the solution passes a collision as an elastic bounce.

Lemma A.2.2 (properties).

- (i) $t(s)$ is C^1 and strictly increasing. The zeros of u, which are the binary collisions, are isolated in s. In particular, collision times do not accumulate in $[t_0, T^*)$.
- (ii) Let $s(t)$ be the inverse function. Then $\boldsymbol{\rho}(t) := \boldsymbol{\rho}(s(t))$ is C^2 on $[t_0, T^*)$, with $\dot{\boldsymbol{\rho}} = \boldsymbol{\sigma} \circ s$ and $\ddot{\boldsymbol{\rho}} = \Phi(\boldsymbol{\rho}, \mathbf{r})$, also at collision times. The function $\mathbf{r}(t) = u(s(t))^2$ is continuous.
- (iii) Between collisions, $(\mathbf{r}, \boldsymbol{\rho})$ solves Newton's equations, and (E) holds with the constant value H.

Proof.

- (i) $t' = |u|^2 \geq 0$. At a zero of u, item 1 of Lemma A.2.1 gives $|u'|^2 = k/2 > 0$. So the zeros of u are isolated, and $t' > 0$ except on a discrete set.
- (ii) The substitution rule for the increasing C^1 map $t(s)$ gives

  $$\boldsymbol{\rho}(s(t)) - \boldsymbol{\rho}(t_0) = \int_{t_0}^{t} \boldsymbol{\sigma}(s(\tau))\, d\tau,$$

  with a continuous integrand. The same argument applied to $\boldsymbol{\sigma}$ gives $\dot{\boldsymbol{\sigma}} = \Phi \circ s$.
- (iii) This is items 3–6 of Lemma A.2.1, together with item 2. $\square$

Proposition A.2.3 (statement (R)). If $\min(r_1, r_2) \geq \delta > 0$ along the continued solution, then $T^* = \infty$.

Proof. Suppose $T^* < \infty$, and put $\Delta = T^* - t_0$. The following bounds hold for all $s \in [0, S^*)$.

- **(B1) Forces.** $|\mathbf{f}| \leq m_0(r_1^{-2} + r_2^{-2}) \leq 2m_0/\delta^2 =: F$, $P \leq 1/\delta$ and $|\Phi| \leq M/\delta^2$. For the last bound: $\Phi = \frac{M}{k}\ddot{\mathbf{x}}_0$ is an algebraic identity in $(\boldsymbol{\rho}, z)$, and $|\ddot{\mathbf{x}}_0| \leq m_1/r_2^2 + m_2/r_1^2 \leq k/\delta^2$.
- **(B2) $\boldsymbol{\sigma}$, $\boldsymbol{\rho}$.** $|\boldsymbol{\sigma}(s) - \boldsymbol{\sigma}(0)| \leq \int |u|^2 |\Phi|\, ds \leq M\Delta/\delta^2$. So $|\boldsymbol{\sigma}| \leq \Sigma := |\dot{\boldsymbol{\rho}}(t_0)| + M\Delta/\delta^2$, and $|\boldsymbol{\rho} - \boldsymbol{\rho}(t_0)| \leq \Sigma\Delta$.
- **(B3) h.** By item 2 of Lemma A.2.1, $mh = H - \frac{1}{2}\mu|\boldsymbol{\sigma}|^2 + \mu MP$. So $|h| \leq A := (|H| + \frac{1}{2}\mu\Sigma^2 + \mu M/\delta)/m$.
- **(B4) u, w.** Write $R = |u|^2$, which is r.
  - Where $R \geq 1$ there is no collision. There $|dR/dt| \leq |\dot{z}| = \sqrt{2h + 2k/R} \leq V := \sqrt{2A + 2k}$.
  - Fix s. Look back from s to the last point at which $R \leq R_0 := \max(1, r(t_0))$, or to $s = 0$ if there is none. In between, $R > R_0 \geq 1$, so the speed bound applies and $R(s) \leq R_0 + V\Delta =: U^2$.
  - By item 1 of Lemma A.2.1, $|w|^2 = \frac{1}{2}(k + h|u|^2) \leq \frac{1}{2}(k + AU^2) =: W_b^2$.

Case $S^* < \infty$.

- By (B2)–(B4), $Y(s)$ stays in a closed and bounded set $\mathcal{K}$ with $r_1, r_2 \geq \delta$.
- $\mathcal{K}$ is compact and contained in D.
- A maximal solution with $S^* < \infty$ must leave every compact subset of D [Ha, Ch. II]. Contradiction.

Case $S^* = \infty$. Then $\int_0^\infty |u|^2 ds = \Delta < \infty$; we show that this integral diverges.

- Choose $\eta \in (0, 1]$ with $A\eta^2 \leq k/2$ and $2\eta + \varepsilon < \sqrt{k}/2$, where $\varepsilon := \frac{A}{2}\eta + \frac{F}{2}\eta^3$.
- *Every interval $[a, a+1]$ contains a point with $|u| > \eta$.* Suppose instead that $|u| \leq \eta$ on all of $[a, a+1]$. Then:
  - $|w'| \leq \varepsilon$ there, so $|w - w(a)| \leq \varepsilon$;
  - $|w(a)|^2 \geq \frac{1}{2}(k - A\eta^2) \geq k/4$;
  - hence $|u(a+1) - u(a)| \geq |w(a)| - \varepsilon \geq \sqrt{k}/2 - \varepsilon > 2\eta$.

  This contradicts $|u(a)|, |u(a+1)| \leq \eta$.
- For $n \geq 1$, take such a point $s^* \in [n, n+1]$.

- Since $|u'| \le W_b$, we have $|u| \ge \eta/2$ on $[s^* - \ell, s^* + \ell] \subset [n-1, n+2]$, where $\ell = \min(\eta/(2W_b), 1)$.
- Hence $\int_{n-1}^{n+2} |u|^2 ds \ge \eta^2 \ell/2 > 0$.

- Summing over $n = 1, 4, 7, \dots$ gives $\int_0^\infty |u|^2 ds = \infty$. Contradiction. $\square$

How A.4 uses this section.

- Step 1 and Step 3 use the continuity of r and $\dot\rho$, which follows from Lemma A.2.2(ii).
- Step 2 uses $\boldsymbol{\rho} \in C^2$ with $\ddot{\boldsymbol{\rho}} = \Phi$, and (E) at times with $r > 0$. These are Lemma A.2.2(ii) and (iii).
- Step 4 uses Proposition A.2.3, which is the statement

(R) If $\inf_{[t_0, T^*)} \min(r_1, r_2) > 0$, then $T^* = \infty$.

Regularity used below. Wherever $\rho > 0$, $\rho = |\boldsymbol{\rho}|$ is C^2 and (RA) holds. This follows from Lemma A.2.2(ii).

A.3 Two inequalities

Lemma A. Let $\lambda \ge \nu > 0$, $\lambda + \nu = 1$, $d \ge 0$ and $\rho > \lambda d$. Then for all $0 \le r \le d$ and all $c \in [-1, 1]$,

$$P(\rho, r, c) \le W(\rho, d).$$

Lemma B. Under the same hypotheses,

$$F(\rho, r, c) \le \hat{F}(\rho, d).$$

Both inequalities become equalities at $(r, c) = (d, -1)$, so they are sharp.

Combining Lemma B with (EM) gives the following. Along a solution, if $\rho > \lambda d$ and $r \le d$, then

$$\frac{\boldsymbol{\rho} \cdot \ddot{\boldsymbol{\rho}}}{\rho} = -MF \;\ge\; -M\hat{F}(\rho, d) = M\,\partial_\rho W(\rho, d). \tag{B$\prime$}$$

Facts on Legendre polynomials P_n.

- (L1) Generating function [DLMF, Eq. 18.12.11]: $(1 - 2xz + z^2)^{-1/2} = \sum_{n \ge 0} P_n(x) z^n$ for $x \in [-1, 1]$ and $|z| < 1$.
- (L2) Laplace's integral [DLMF, Eq. 18.10.5]: $P_n(\cos\vartheta) = \frac{1}{\pi} \int_0^\pi (\cos\vartheta + i \sin\vartheta \cos\phi)^n \, d\phi$. Since $|\cos\vartheta + i\sin\vartheta\cos\phi|^2 = \cos^2\vartheta + \sin^2\vartheta\cos^2\phi \le 1$, we get $|P_n(x)| \le 1$ on $[-1, 1]$.
- (L3) In (L1), replacing (x, z) by $(-x, -z)$ gives $P_n(-x) = (-1)^n P_n(x)$. Setting $x = 1$ gives $\sum P_n(1) z^n = (1 - z)^{-1}$, so $P_n(1) = 1$. Hence $P_n(-1) = (-1)^n$.

Proof of Lemmas A and B.

Step 1 (normalization).

- Put $s = r/\rho$, $a = \nu s$ and $b = \lambda s$. Since $\lambda r \le \lambda d < \rho$, we have $0 \le a \le b < 1$.
- Now $r_1 = \rho\sqrt{1 - 2ac + a^2}$ and $\rho - \nu r c = \rho(1 - ac)$.
- Similarly $r_2 = \rho\sqrt{1 - 2b(-c) + b^2}$ and $\rho + \lambda r c = \rho(1 - b(-c))$.
- Therefore

$$\rho P = \lambda G(a, c) + \nu G(b, -c), \qquad \rho^2 F = \lambda g(a, c) + \nu g(b, -c),$$

  where

$$G(a, x) = (1 - 2ax + a^2)^{-1/2}, \qquad g(a, x) = (1 - ax)(1 - 2ax + a^2)^{-3/2}.$$

Step 2 (expansions).

- By (L1), $G(a, x) = \sum_n a^n P_n(x)$.
- One checks directly that $g = G + a\,\partial_a G$. Hence $g(a, x) = \sum_n (n+1) a^n P_n(x)$.
- For $0 \le a < 1$ and $|x| \le 1$ both series converge absolutely, since $\sum (n+1) a^n |P_n(x)| \le (1 - a)^{-2}$ by (L2). So they may be added and regrouped term by term.

Step 3 (combination). By Steps 1 and 2 together with (L3),

$$\rho^2 F = \sum_{n\ge 0} (n+1) s^n \beta_n P_n(c), \qquad \rho P = \sum_{n\ge 0} s^n \beta_n P_n(c), \qquad \beta_n := \lambda\nu^n + \nu(-\lambda)^n.$$

- $\beta_0 = 1$, and $\beta_1 = \lambda\nu - \nu\lambda = 0$ (no dipole term about $\mathbf{g}$).
- **For every n, $(-1)^n\beta_n \ge 0$.**
  - For even n, $(-1)^n\beta_n = \lambda\nu^n + \nu\lambda^n > 0$.
  - For odd n, $(-1)^n\beta_n = \nu\lambda^n - \lambda\nu^n = \lambda\nu(\lambda^{n-1} - \nu^{n-1}) \ge 0$, because $\lambda \ge \nu$.
  - This is the only place where $m_1 \le m_2$ is used.
- Consequently $|\beta_n| = (-1)^n\beta_n$.

Step 4 (maximum over the angle). By (L2) and Step 3,

$$\rho^2 F \le \sum_n (n+1) s^n |\beta_n| = \lambda \sum_n (n+1)(-\nu s)^n + \nu \sum_n (n+1)(\lambda s)^n = \frac{\lambda}{(1+\nu s)^2} + \frac{\nu}{(1-\lambda s)^2}.$$

In the same way,

$$\rho P \le \sum_n s^n |\beta_n| = \frac{\lambda}{1+\nu s} + \frac{\nu}{1-\lambda s}.$$

The right-hand sides equal $\rho^2 F(\rho, r, -1)$ and $\rho P(\rho, r, -1)$, respectively.

Step 5 (maximum over the separation).

- Each term $(n+1)s^n|\beta_n|$ and $s^n|\beta_n|$ is nondecreasing in $s \ge 0$.
- Since $r \le d$, we have $s \le s_d := d/\rho < 1/\lambda$, so all series still converge at s_d.
- Therefore

$$\rho^2 F \le \frac{\lambda}{(1+\nu s_d)^2} + \frac{\nu}{(1-\lambda s_d)^2} = \rho^2 \hat F(\rho, d), \qquad \rho P \le \frac{\lambda}{1+\nu s_d} + \frac{\nu}{1-\lambda s_d} = \rho\, W(\rho, d).$$

  Dividing by ρ^2 and by ρ gives the lemmas. □

Remarks on the range.

- (i) No upper bound on ρ is needed.
- (ii) The bounds blow up as $\rho \to \lambda d^+$, but they hold at every point with $\rho > \lambda d$. No uniformity is used.
- (iii) $r = 0$ is included; this covers the instant of a binary collision.
- (iv) Only $c = \cos\theta$ enters, so the lemmas hold in the plane and in space.
- (v) If $\lambda = \nu$, all odd β_n vanish and equality also holds at $c = +1$.

Remark (second proof of Lemma A). Fix r. Then:

- $\partial_c P = \lambda\nu\rho r\,(r_1^{-3} - r_2^{-3})$, and $r_2^2 - r_1^2 = r\big((\lambda-\nu)r + 2\rho c\big)$.
- So P decreases in c up to $c^* = -(\lambda-\nu)r/(2\rho)$ and increases after it. Its maximum over $[-1,1]$ is therefore attained at $c = \pm 1$.
- $P(\rho, r, -1) - P(\rho, r, 1) = 2\lambda\nu r\big[(\rho^2 - \lambda^2 r^2)^{-1} - (\rho^2 - \nu^2 r^2)^{-1}\big] \ge 0$.
- Finally, $P(\rho, r, -1) = W(\rho, r)$ and $\partial_d W = \lambda\nu\big[(\rho - \lambda d)^{-2} - (\rho + \nu d)^{-2}\big] \ge 0$.

A.4 The criterion

Theorem (★). Let $m_1 \le m_2$ and $H < 0$, in the barycentric frame. Suppose that at some time t_0

- **(C1)** $r(t_0) < d_1 = m_1 m_2 / K$,
- **(C2)** $\rho(t_0) > \lambda d_1$,
- **(C3)** $\dot\rho(t_0) > \sqrt{2M\, W(\rho(t_0), d_1)}$.

Let $C := \sqrt{\dot\rho(t_0)^2 - 2M\,W(\rho(t_0), d_1)} > 0$. Then the solution, continued through binary collisions as in A.2, exists for all $t \geq t_0$. Moreover, for all $t \geq t_0$:

1. $r(t) \leq \dfrac{m_1 m_2}{K + \frac{1}{2}\mu C^2} < d_1$;
2. $\frac{1}{2}\dot\rho(t)^2 - M\,W(\rho(t), d_1) \geq \frac{1}{2}C^2$ and $\dot\rho(t) > C$;
3. $\rho(t) \geq \rho(t_0) + C\,(t - t_0)$.

In particular $\rho \to \infty$ while the binary separation stays bounded; that is, the motion is hyperbolic–elliptic.

Proof.

Step 1 (bootstrap set). Let

$$\mathcal{G} = \big\{ t \in [t_0, T^*) :\ r(s) \leq d_1 \text{ and } \dot\rho(s) \geq 0 \text{ for all } s \in [t_0, t] \big\}, \qquad \tau = \sup \mathcal{G}.$$

$\mathcal{G}$ is an interval containing t_0. By (C1) and (C3), $r(t_0) < d_1$ and $\dot\rho(t_0) > 0$, and both r and $\dot\rho$ are continuous. Hence $\tau > t_0$.

Step 2 (estimates on $\mathcal{G}$). Fix $t \in \mathcal{G}$.

(a) Separation.

- ρ is nondecreasing on $[t_0, t]$. So for $s \in [t_0, t]$ we have $\rho(s) \geq \rho(t_0) > \lambda d_1 \geq \lambda r(s)$.
- Put $\delta := \rho(t_0) - \lambda d_1 > 0$. Then on $[t_0, t]$

$$r_2 \geq \rho - \lambda r \geq \delta, \qquad r_1 \geq \rho - \nu r \geq \rho(t_0) - \nu d_1 \geq \delta. \tag{SEP}$$

(b) Radial velocity.

- For $s \in [t_0, t]$, (RA) and (B′) with $d = d_1$ give $\ddot\rho(s) \geq M\,\partial_\rho W(\rho(s), d_1)$.
- Multiplying by $\dot\rho(s) \geq 0$ gives $\dot\rho\ddot\rho \geq \frac{d}{ds}\big[M\,W(\rho(s), d_1)\big]$.
- Integrating (this uses $\rho \in C^2$, see A.2):

$$\tfrac{1}{2}\dot\rho(t)^2 - M\,W(\rho(t), d_1) \ \geq\ \tfrac{1}{2}\dot\rho(t_0)^2 - M\,W(\rho(t_0), d_1) = \tfrac{1}{2}C^2.$$

- Since $W > 0$ and $\dot\rho(t) \geq 0$, it follows that $\dot\rho(t) > C$.

(c) Binary separation.

- If $r(t) = 0$ there is nothing to prove.
- Otherwise use (E), $\frac{1}{2}m|\dot{\mathbf{r}}|^2 \geq 0$, $|\dot{\boldsymbol{\rho}}| \geq |\dot\rho|$, Lemma A (valid since $r(t) \leq d_1$ and $\rho(t) > \lambda d_1$), and (b):

$$\frac{m_1 m_2}{r(t)} = K + \tfrac{1}{2}m|\dot{\mathbf{r}}|^2 + \tfrac{1}{2}\mu|\dot{\boldsymbol{\rho}}|^2 - \mu MP \ \geq\ K + \mu\Big(\tfrac{1}{2}\dot\rho^2 - M\,W(\rho, d_1)\Big) \ \geq\ K + \tfrac{1}{2}\mu C^2.$$

- Hence $r(t) \leq m_1 m_2/(K + \frac{1}{2}\mu C^2) < d_1$.

Step 3 ($\tau = T^*$). Suppose $\tau < T^*$.

- By continuity, $r(\tau) \leq d_1$ and $\dot\rho(\tau) \geq 0$. Hence $\tau \in \mathcal{G}$.
- By Step 2, $r(\tau) < d_1$ and $\dot\rho(\tau) > C > 0$.
- By continuity these strict inequalities persist on $[\tau, \tau + \varepsilon)$ for some $\varepsilon > 0$. So $\tau + \varepsilon/2 \in \mathcal{G}$, which contradicts $\tau = \sup \mathcal{G}$.
- Therefore $\mathcal{G} = [t_0, T^*)$.

Step 4 (global existence). By (SEP), $\min(r_1, r_2) \geq \delta > 0$ on $[t_0, T^*)$. So $T^* = \infty$ by (R).

Step 5 (conclusion). Items 1 and 2 hold for every $t \geq t_0$ by Steps 2–4. Item 3 follows by integrating $\dot\rho > C$. □

Remarks

A.4.1 (angular momentum). The only place where information is discarded is (RA), and the discarded term is nonnegative. The theorem therefore applies to $L = 0$, which is the case of the Pythagorean problem.

A.4.2 (relation to Yoshida's statements). These relations are based on the statements of [Y72], [Y73] and [Y74] as transcribed from the original scans.

Theorem 1 of [Y72]. Its hypotheses are:

- (13) $\rho(0) \geq ad > a\,r(0)$ for some $a > \lambda$;
- (14) $\dot\rho(0) > \sqrt{2MW(\rho(0), d)}$;
- (15) $\dot\rho(0)^2 - 2MW(\rho(0), d) > 2H/\mu + 2m_1m_2/(\mu d)$.

For $d = d_1$:

- The right side of (15) is 0, so (15) follows from (14).
- (13) is equivalent to (C1) together with (C2): take $a \in (\lambda, \rho(t_0)/d_1]$.

So (★) is Theorem 1 of [Y72] with $d = m_1m_2/K$.

Criterion A′ of [Y74]. With $d = d_1$ it requires $\rho(0) \geq ad_1 \geq a\,r(0)$, which allows the boundary case $r(0) = d_1$. That case cannot occur together with (C2) and (C3): Step 2(c) at $t = t_0$ would force $r(t_0) < d_1$.

Corollary of [Y72]. It has the gap described in A.0, which was corrected in [Y73] and again in [Y74, p. 369]. We do not use it.

A.4.3 (ordering of the masses). If $m_1 > m_2$, relabel the binary components. With λ and ν interchanged in W, the function becomes smaller, and Lemmas A–B fail, for example at $(r, c) = (d, -1)$. The ordering is essential.

A.4.4 (checkability). (C1)–(C3) are strict inequalities in $(r, \rho, \dot\rho)$ with exact constants. It suffices to check them for every state in a rigorous enclosure of the true state at a single time t_0. The value of K must be the exact energy of the true solution, not the energy recomputed from an enclosure.

A.4.5 (non-accumulation of collisions). Version 1 gave here only a sketch of why binary collisions cannot accumulate under (SEP). In version 1.2 this is proved: collisions are isolated by Lemma A.2.2(i), and the solution exists for all time by Proposition A.2.3.

A.5 Application to the Pythagorean problem

Initial data [Bu1913; SP1967]. Masses $3, 4, 5$ start at rest at $(1, 3)$, $(-2, -1)$ and $(1, -1)$, respectively.

- The barycentre is at the origin, and the total momentum and angular momentum vanish.
- The mutual distances are $|3\text{–}4| = 5$, $|3\text{–}5| = 4$ and $|4\text{–}5| = 3$.
- Hence, exactly,

$$H = -\Big(\frac{3\cdot 4}{5} + \frac{3\cdot 5}{4} + \frac{4\cdot 5}{3}\Big) = -\frac{769}{60}, \qquad K = \frac{769}{60}.$$

- The energy is conserved along the (regularized) solution, so this K is the value to use at any later time.

Parameters. The escaper is $m_0 = 3$, and the binary is $m_1 = 4 \leq m_2 = 5$:

$$M = 12, \quad \lambda = \tfrac{5}{9}, \quad \nu = \tfrac{4}{9}, \quad \mu = \tfrac{9}{4}, \quad m = \tfrac{20}{9}, \quad d_1 = \frac{m_1m_2}{K} = \frac{1200}{769} \approx 1.560468140442,$$

$$\lambda d_1 = \frac{2000}{2307} \approx 0.866926744690, \qquad \nu d_1 = \frac{1600}{2307} \approx 0.693541395752.$$

Conditions. Let $\mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5$ be the positions of the masses $3, 4, 5$. Then

$$r = |\mathbf{x}_5 - \mathbf{x}_4|, \qquad \boldsymbol{\rho} = \mathbf{x}_3 - \frac{4\mathbf{x}_4 + 5\mathbf{x}_5}{9}, \qquad \dot\rho = \frac{\boldsymbol{\rho}\cdot\dot{\boldsymbol{\rho}}}{\rho},$$

and the conditions of Theorem (★) read

$$\text{(C1) } r < \frac{1200}{769}, \qquad \text{(C2) } \rho > \frac{2000}{2307}, \qquad \text{(C3) } \dot\rho > \sqrt{24\,W(\rho)}, \quad W(\rho) = \frac{5/9}{\rho + 1600/2307} + \frac{4/9}{\rho - 2000/2307}.$$

In W, the coefficient $5/9$ goes with the **larger** distance $\rho + \nu d_1$, and $4/9$ with the **smaller** distance $\rho - \lambda d_1$.

Conclusion. If (C1)–(C3) hold for the true state at some t_0, then for all $t \geq t_0$:

- the 4–5 separation stays below $m_1 m_2/(K + \frac{1}{2}\mu C^2) < 1200/769$;
- the body of mass 3 recedes at least linearly: $\rho(t) \geq \rho(t_0) + C(t - t_0)$.

Here $\mu = 9/4$ and C is defined from the state at t_0.

Verifying (C1)–(C3) for the actual trajectory is a separate, computer-assisted step described in the main text. It is not part of this appendix.

Historical note. Yoshida [Y72, p. 406] already applied the corollary to this problem at $t = 59.72$, using an earlier non-rigorous numerical solution. At that point $r \approx 0.42 < d_1$, so the corrected condition was also satisfied.

A.6 Independent cross-checks (not part of the proof)

These checks look for errors in the proof above; they do not replace it.

- **Computer algebra (SymPy).** We checked every algebraic identity used in A.1 and A.3: (E) and (EM) derived from the Cartesian Newton equations, (RA) in 2D and 3D, the expansions of Steps 1–3 up to finite order, the closed forms of Step 4, the second proof of Lemma A, and, in version 1.2, the 10 identities of Lemma A.2.1 (`proof_review/s3/s3_algebra_check.py`). Deliberately wrong variants were rejected.
- **Regularized vs. Cartesian integration (A.2, non-rigorous).** For the Pythagorean data, the system (LC) integrated in s and the Newton equations integrated in t agree at $t = 30$ to 7.5×10^{-8} (binary separation) and 2.1×10^{-8} ($\boldsymbol{\rho}$), through a 4–5 approach of 0.0022; the invariant I drifts by 2×10^{-12} (`proof_review/s3/s3_numeric_sanity.py`).
- **Adversarial numerical search** (50-digit arithmetic, 40 000 points concentrated near the boundaries $\rho \to \lambda d^+$, $r \in \{0, d\}$, $c = \pm 1$, for several $\lambda \in [\frac{1}{2}, 1)$). We found no violation of Lemmas A–B. The maximum ratio was $1 + O(10^{-50})$, attained at $(r, c) = (d, -1)$.
- **Dynamical falsification test.** We took 800 states with masses $3, 4, 5$ and various energies satisfying (C1)–(C3), and integrated them to $t \leq 40$. No conclusion of Theorem (★) was violated. Variants with the factor 2 in (C3) omitted, with (C1) dropped, or with λ, ν interchanged in W did produce violations, including the escaper returning.

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